

Machine Learning & Statistics - Interview Ready Notes



Perfect balance: Concepts clear, examples ready, interview-friendly

1. Z-SCORE

What is it? Z-score batata hai ki koi value mean se kitne standard deviations door hai.

Formula:

$$Z = (X - \mu) / \sigma$$

Example:

```
Class marks: Mean = 70, SD = 10
Your marks = 85
Z = (85-70)/10 = 1.5
Matlab: You're 1.5 SD above average (top ~7%)
```

When to use:

- Outlier detection ($|Z| > 3$ = outlier)
- Comparing different scales
- Standardization

Interview Tip: "Z-score is a way to standardize data so we can compare values from different distributions on the same scale."

2. NORMAL DISTRIBUTION

What is it? Bell-shaped curve where most data clusters around mean.

Key Rules (68-95-99.7):

- 68% data within 1 SD
- 95% data within 2 SD
- 99.7% data within 3 SD

Connection with Z-score: Z-score tells you where a point sits on the normal curve.

Real examples:

- Heights, IQ scores, test marks

- Many natural phenomena

Interview Question: "Why is normal distribution important?" **Answer:** "Many statistical tests assume normality, and due to Central Limit Theorem, sample means tend to be normally distributed even if original data isn't."

3. STANDARDIZATION

What: Transform data to Mean = 0, SD = 1

Formula: Same as Z-score: $(X - \mu) / \sigma$

When to use: ☒ Features on different scales (age vs salary) ☒ Distance-based algorithms (KNN, SVM) ☒
Data roughly normal

Example:

```
Before: Height=[150,180] Weight=[50,80]
After: Both have mean=0, sd=1
Now fair comparison!
```

Interview Tip: "Standardization maintains the shape of distribution but centers it at 0 with unit variance."

4. NORMALIZATION

What: Scale data to [0,1] range

Formula: $(X - \min) / (\max - \min)$

When to use: ☒ Bounded data (0-100 marks) ☒ Neural networks ☒ When you don't assume normal distribution

Example:

```
Marks: [20, 40, 60, 80, 100]
Normalized: [0, 0.25, 0.5, 0.75, 1.0]
```

5. STANDARDIZATION vs NORMALIZATION

Aspect	Normalization	Standardization
Range	[0, 1]	Unbounded
Formula	$(X - \min) / (\max - \min)$	$(X - \text{mean}) / \text{SD}$
Mean/SD	Not 0/1	Always 0/1

Aspect	Normalization	Standardization
Use when	Bounded data, NNs	Normal data, distance algos
Outlier sensitivity	High	Lower

Interview Answer: "Choose normalization for bounded data like images (0-255 pixels). Choose standardization when data is normally distributed and you're using algorithms sensitive to scale like SVM or KNN."

6. LABELLED vs UNLABELLED DATA

Labelled Data

- **What:** Input + Output both given
- **Learning:** Supervised
- **Example:** Email → Spam/Not Spam
- **Use:** Classification, Regression

Unlabelled Data

- **What:** Only Input, no output
- **Learning:** Unsupervised
- **Example:** Customer grouping
- **Use:** Clustering, Pattern discovery

Interview Question: "Which is more common in real world?" **Answer:** "Unlabelled data is abundant and cheap. Labelled data is expensive (requires human labeling) but more useful for prediction tasks."

7. IMPORTANT ML TERMS

Term	Simple Meaning	Example
Feature	Input variable	Age, Salary
Label	What we predict	Price, Category
Training	Data for learning	80% of dataset
Testing	Data for evaluation	20% of dataset
Model	Learner (predictor)	Linear Regression
Algorithm	Learning method	Gradient Descent
Overfitting	Memorizing training data	98% train, 60% test
Underfitting	Too simple model	60% train, 58% test

Overfitting vs Underfitting:

Underfitting: Model too simple, doesn't learn
Balanced: Good generalization
Overfitting: Model too complex, memorizes noise



8. LOG TRANSFORMATION

Why use it? Compress large values, reduce outlier effect

When to use:

- Right-skewed data
- Values span multiple magnitudes
- Income, population, prices

Example:

Income: [20K, 30K, 40K, 500K] ← 500K is outlier
Log: [4.3, 4.5, 4.6, 5.7] ← More balanced

Interview Tip: "Log transformation makes multiplicative relationships additive and helps normalize skewed data."

△ 9. TRIANGULATION

What: Study one thing from multiple angles for stronger conclusions

4 Types:

1. **Data Triangulation:** Different data sources
 - Example: Hospital records + surveys + sensors
2. **Method Triangulation:** Different research methods
 - Example: Surveys + interviews + observations
3. **Theory Triangulation:** Different theoretical perspectives
 - Example: Psychology + sociology + economics
4. **Investigator Triangulation:** Multiple researchers
 - Example: 3 doctors diagnose same X-ray

Why important:

- Reduces bias
- Increases credibility

- Comprehensive understanding

Interview Answer: "Triangulation is like getting a second opinion - when multiple independent sources/methods confirm the same finding, we can be more confident it's true."

10. SURROGATE ENDPOINTS

What: Short-term measure that predicts long-term outcome

Why use:

- Faster results (months vs years)
- Cheaper
- More ethical

Example:

```
Real outcome: Heart attack (takes years to observe)
Surrogate: Blood pressure (measure in weeks)
Logic: Lower BP → Lower heart attack risk
```

Risk: Surrogate might not always predict true outcome correctly

Interview Question: "Give an example where surrogate endpoint failed" **Answer:** "Some drugs lowered cholesterol (surrogate) but didn't reduce heart attacks (true outcome). This shows importance of validating surrogates."

11. TRUE & ERROR SCORES

Formula: $\text{Observed Score} = \text{True Score} + \text{Error}$

Example:

```
Your true ability = 75
Test 1: 70 (nervous, -5 error)
Test 2: 80 (lucky guesses, +5 error)
Test 3: 75 (normal day, 0 error)

Average observed = 75 ≈ True score
```

Random Error

- Unpredictable, different each time
- Cancels out over multiple measures
- Example: Lucky guesses, momentary distraction

Systematic Error (Bias)

- Consistent, same direction
- Does NOT cancel out
- Example: Scale always +2kg, broken thermometer

Interview Tip: "Random errors affect precision (consistency), systematic errors affect accuracy (correctness)."

⚠ 12. TYPE I & TYPE II ERRORS

Setup: Testing if new drug works

Reality	Decision	Result
Drug doesn't work	Reject H_0 (say it works)	Type I Error (False Positive)
Drug works	Accept H_0 (say it doesn't)	Type II Error (False Negative)

Memory Trick:

- **Type I:** Fire alarm WITHOUT fire (false alarm)
- **Type II:** Fire WITHOUT alarm (missed detection)

In ML:

- Type I = False Positive
- Type II = False Negative

Trade-off:

Reduce Type I $\rightarrow$ Type II increases
 Example: Stricter spam filter

- Less false positives (important emails marked spam)
- More false negatives (spam in inbox)

Interview Question: "Which error is worse?" **Answer:** "Depends on context. In cancer screening, Type II (missing disease) is worse. In spam filtering, Type I (blocking important email) is worse."

☑ 13. RELIABILITY vs VALIDITY

Reliability = Consistency

"Same test, same person, multiple times $\rightarrow$ similar scores?"

Types:

1. **Test-Retest:** Same test twice
2. **Inter-Rater:** Two observers agree
3. **Internal Consistency:** All questions measure same thing (Cronbach's α)

Example:

Weighing scale shows: 70, 70, 70, 70 kg
→ Highly reliable (consistent)

Validity = Accuracy

"Does test measure what it claims?"

Types:

1. **Content Validity:** Covers full range
 - Example: Driving test includes parking + highway + rules
2. **Criterion Validity:** Predicts related outcome
 - Example: SAT scores predict college GPA
3. **Construct Validity:** Measures theoretical concept
 - Example: Depression test correlates with therapy outcomes

Example:

Scale always shows 72 kg (you're actually 70 kg)
→ Reliable but NOT valid

Dart Board Analogy:

Reliable + Valid: Arrows clustered at bullseye ✓
Reliable only: Arrows clustered but off-center
Neither: Arrows scattered everywhere

Interview Key Point: "You can be reliable without being valid, but you cannot be valid without being reliable."

14. MEASUREMENT BIAS

What: Systematic error in data collection

Common Types:

1. Sampling Bias

- Sample ≠ Population

- Example: Online survey misses non-internet users

2. Selection Bias

- How you select affects results
- Example: Only volunteers participate (healthier people)

3. Confirmation Bias

- See only what confirms beliefs
- Example: Cherry-picking data that supports hypothesis

4. Survivorship Bias

- Only see successes, ignore failures
- Example: Study successful startups, ignore failed ones

5. Reporting Bias

- Selective reporting
- Example: Publish positive results, hide negative

How to Minimize:

- Random sampling
- Blinding (researchers don't know groups)
- Multiple data sources
- Pre-register hypotheses

Interview Story: "WW2 plane example - Engineers wanted to reinforce areas with bullet holes (where returning planes were hit). Statistician said reinforce areas WITHOUT holes - because planes hit there didn't return! Classic survivorship bias."

15. STATISTICS TYPES

Descriptive Statistics

Purpose: Summarize data

Key measures:

- Mean, Median, Mode (central tendency)
- SD, Variance, Range (spread)
- Skewness, Kurtosis (shape)

Example: "Average class score is 75 with SD of 10"

Inferential Statistics

Purpose: Population predictions from sample

Key concepts:

- Hypothesis testing (t-test, ANOVA, chi-square)
- Confidence intervals
- p-values

Example: "Based on 1000 voters, we're 95% confident the candidate has 52-58% support"

Interview Answer: "Descriptive tells us what happened in our data. Inferential helps us generalize to broader population."

16. DATA ANALYSIS TYPES

Type	Question	Example	Use
Descriptive	What happened?	Sales summary	Reports
Diagnostic	Why happened?	Why sales dropped?	Root cause
Predictive	What will happen?	Next month sales	Forecasting
Prescriptive	What should we do?	Best marketing strategy	Optimization
Exploratory (EDA)	What patterns exist?	Find clusters	Initial analysis
Inferential	Can we generalize?	Sample → Population	Research

Interview Flow: "I'd start with EDA to understand data, then descriptive to summarize, diagnostic to find causes, predictive to forecast, and finally prescriptive to recommend actions."

QUICK INTERVIEW TIPS

Common Questions & Answers:

Q: Difference between normalization and standardization? A: "Normalization scales to [0,1], good for bounded data and neural networks. Standardization centers at mean=0 with SD=1, good for normally distributed data and distance-based algorithms."

Q: How to handle outliers? A: "First investigate - is it error or real? If real: use robust methods (median instead of mean), log transform, or use algorithms robust to outliers like tree-based models."

Q: What's overfitting and how to prevent? A: "Model memorizes training data instead of learning patterns. Prevent by: more data, simpler model, regularization (L1/L2), cross-validation, or early stopping."

Q: Explain bias-variance tradeoff A: "High bias = underfitting (too simple). High variance = overfitting (too complex). Goal is to find sweet spot - model complex enough to capture patterns but simple enough to generalize."

Q: When would you use supervised vs unsupervised learning? A: "Supervised when you have labeled data and clear prediction goal (spam detection, price prediction). Unsupervised when exploring unlabeled data for patterns (customer segmentation, anomaly detection)."

CONCEPT CONNECTIONS

How concepts relate:

```
Raw Data
  ↓
EDA (Explore patterns)
  ↓
Preprocessing:
- Handle outliers (log transform, Z-score detection)
- Scale features (normalize/standardize)
- Check quality (reliability, validity)
- Reduce bias (sampling, measurement)
  ↓
Split: Train/Test
  ↓
Model Training (avoid overfitting/underfitting)
  ↓
Evaluation (check Type I/II errors)
  ↓
Statistical Analysis (descriptive, inferential)
  ↓
Results (triangulation for validation)
```

FINAL REVISION CHECKLIST

Before interview, make sure you can explain:

☒ Z-score formula and when to use it ☒ Difference between normalization & standardization ☒ Labelled vs unlabelled data examples ☒ Overfitting vs underfitting with examples ☒ Type I vs Type II errors with real examples ☒ Reliability vs Validity with dart board analogy ☒ At least 2 types of bias with examples ☒ Descriptive vs Inferential statistics ☒ All 4 types of data analysis with use cases ☒ How to handle outliers (log transform)

Practice Statement: "I understand the fundamentals of data preprocessing, statistical analysis, and machine learning concepts. I can explain trade-offs, identify biases, and choose appropriate methods based on data characteristics and business needs."

REMEMBER

Key Philosophy:

- **No perfect model** - always trade-offs
- **Garbage in, garbage out** - data quality matters most
- **Context is king** - right method depends on situation
- **Validate everything** - use multiple methods (triangulation)
- **Question assumptions** - check for biases

Interview Mindset: Don't just memorize - **understand WHY**. Interviewers want to see you can:

1. Think critically
2. Choose right approach
3. Explain trade-offs
4. Connect concepts
5. Apply to real problems

Good luck! 🚀